Class: Grade 12 LS **Subject: Mathematics**

Teacher: Muhieddeen Iskandarani **Course: inverse function**

I.Existence of inverse function denoted by f^{-1} .

f admits an inverse function over the interval I = [a; b] denoted byf^{-1} **if and only if** f is continuous and strictly monotonic over I.

The inverse function f^{-1} is defined by: if $f(x) = y$ then $x = f^{-1}(y)$.

II.Domain of the inverse function f^{-1}

If $I = [a, b]$ is the domain of definition of f, then **the domain of $f^{-1}$is image of** $[a, b]$ by f:

$\boxed{D_{f^{-1}} = f(I)}$

If $f$is **strictly increasing** on I, then:$\boxed{D_{f^{-1}} = [f(a), f(b)]}$

If $f$is **strictly decreasing** on I, then:$\boxed{D_{f^{-1}} = [f(b), f(a)]}$

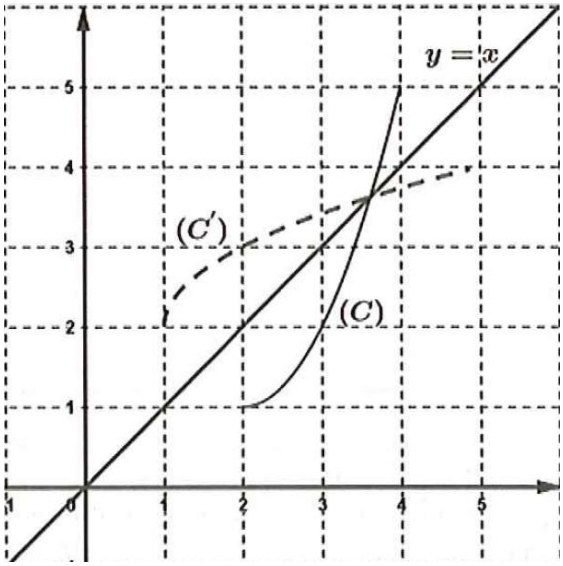
III.Curve of f^{-1}:

Designate by $(C)$and $(C')$the representative curves of $f$and f^{-1}respectively.

Then $(C')$is the symmetric of $(C)$with respect to the first bisector $y = x$.

Comparison of the two curves $(C)$and (C'):

curve (C)	**Curve** (C')
$x = a$ **V.A**	$y = a$ **H. A**
$y = b$ **H. A**	$x = b$**V.A**
$y = ax + b$ **O. A**	$x = ay + b$ **O. A**
Slope of tangent $(T)$to (C)at a point $M(a; b)$is m.	Slope of tangent $(T')$to (C')at a point $M'(b; a)$is $\frac{1}{m}$.



Remark:

$(C)$and (C')intersect each other at the straight line $y = x$.

IV.Explicit Form of f^{-1}

$y = f(x) \Leftrightarrow x = f^{-1}(y)$

To find the explicit form of f^{-1}, follow these steps:

1. Let $y = f(x)$
2. Find x as a function of y that is, $x = f^{-1}(y)$.
3. Swap $x$and y, replace $x$by $y$and $y$by x.

Exercise:

Determine f^{-1}in each of the following cases

1. $f(x) = x^2, I = [0 + \infty[$,
2. $f(x) = \frac{x+1}{2x-1}, I = \mathbb{R} - \left\{\frac{1}{2}\right\}$
3. $f(x) = (x-3)^2, I =]-\infty, 3]$

Solutions: Finding f^{-1}

1. $y = x^2 \Rightarrow x = \pm\sqrt{y}$. Since $x > 0$(domain $I = [0 + \infty[$), $x = \sqrt{y} = f^{-1}(y)$

   Change y into x then $f^{-1}(x) = \sqrt{x}$.

2. $y = \frac{x+1}{2x-1} \Rightarrow 2xy - y = x + 1 \Rightarrow x(2y-1) = y + 1 \Rightarrow x = \frac{y+1}{2y-1}$.

   Hence, $f^{-1}(y) = \frac{y+1}{2y-1}$.

   Change y into x then $f^{-1}(x) = \frac{x+1}{2x-1}$

3. $y = (x-3)^2 \Rightarrow x - 3 = \pm\sqrt{y} \Rightarrow x = 3 \pm \sqrt{y}$.Since $x \leq 3$(domain $I =]-\infty, 3]$),

   $x = 3 - \sqrt{y} \Rightarrow f^{-1}(y) = 3 - \sqrt{y}$.

   change y into x then $f^{-1}(x) = 3 - \sqrt{x}$

V.Derivative of f^{-1}

Let $f(x)$be a differentiable function on an interval I. If $f'(x) \neq 0$and if $f(x_0) = y_0$, then $f^{-1}(y_0) = x_0$, and:

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)} = \frac{1}{f'\left(f^{-1}(y_0)\right)}$$

Remark: How to calculate $(f^{-1})'(a)$?

1. Solve the equation $f(x) = a$to find $x = x_0$.
2. Then write:$(f^{-1})'(a) = \frac{1}{f'(x_0)}$

Application Example

Problem: For $f(x) = x^3 + 2x + 4$, calculate $(f^{-1})'(4)$.

Solution:

$f(x) = 4 \Rightarrow x^3 + 2x + 4 = 4 \Rightarrow x(x^2 + 2) = 0 \Rightarrow x = 0$

So,$(f^{-1})'(4) = \frac{1}{f'(0)} = \frac{1}{2}$

VI.Intersection between C_f and $C_{f^{-1}}$

To find the coordinates of the point of intersection between a function $f$and its inverse f^{-1}(when it exists), we solve the equation: $f(x) = x$

Application Example

Problem: Let f be the function defined over $]0 + \infty[$; by $f(x) = \frac{1}{x}$. Determine the coordinates of the point of intersection between $f$and its inverse f^{-1}.

Solution:

$f'(x) = \frac{-1}{x^2} < 0 \Rightarrow f$ is strictly decreasing over $\left]0; +\infty\right[$

Since $f$is also continuous over $]0 + \infty[$;, it admits an inverse function f^{-1}.

Now solve:$f(x) = x \Rightarrow \frac{1}{x} = x \Rightarrow x^2 = 1 \Rightarrow x = 1$ or $x = -1$

But $x = -1$ is rejected (outside domain). Thus, the intersection point is:$(1, 1)$